



Spatial genetic structure at the leading edge of a spruce budworm outbreak: The role of dispersal in outbreak spread

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ABSTRACT

Outbreking insects often undergo rapid population growth synchronously in multiple locations separated by large distances. Dispersal may play an essential role in synchronizing outbreaks over large geographic areas but its role in outbreak spread remains unclear. In our study, we used population genetics to assess how much dispersal contributes to the spread of a major forest pest, the spruce budworm (*Choristoneura fumiferana* Clemens). Using spatial analyses of genotypic data, we determined the extent to which dispersing individuals from epidemic populations in Quebec leave their genetic signature in adjacent, rising populations in New Brunswick and Maine. A lack of genetic differentiation between endemic and epidemic populations would indicate that outbreak spread depends on effective dispersal from epidemic to endemic regions. In contrast, endemic populations that are distinct from epidemic population would suggest that dispersal plays a reduced role in fomenting outbreak rise. An intermediate possibility with differentiation between epidemic and endemic population as a continuous spatial gradient might suggest a temporal lag in genetic differentiation that may take the form of a genetic travelling wave. PCA, cluster analyses, isolation by distance, and sPCA were used to characterize spatial genomic variation using 300 SBW larvae sampled in 2015 and genotyped at thousands of genome-wide SNPs. We found support for a genetic travelling wave pattern, matching the SBW density pattern. These results indicate that dispersal may trigger the transition of populations from the endemic to the epidemic state and thereby drive outbreak spread. By supplementing persisting endemic populations, migrants may drive populations to beyond levels that can be controlled by local biotic constraints. These results support the underlying justification of ongoing outbreak containment efforts in Atlantic Canada (*i.e.*, the Early Intervention Strategy), although continued sampling as the outbreak progresses is needed confirm the temporal stability of the observed patterns.

1. Introduction

Insect outbreaks are a common disturbance in forest ecosystems and occur both periodically and synchronously over large geographic areas (Myers, 1993). This pattern has been documented for a variety of major insect defoliators in North America, including the forest tent caterpillar (*Malacosoma disstria*) and the jack pine budworm (*Choristoneura pinus pinus*). When outbreking, these insects can affect millions of hectares of susceptible forest (Cooke et al., 2007). Even though outbreaks of these species and their consequences have been well studied, there remains uncertainty around the underlying processes, with particular debate around the role dispersal plays in synchronizing vs. spreading

outbreaks (Royama, 1984; Régnière and Nealis, 2019). This debate has implications for our understanding of how and why outbreaks develop, and in turn what strategic management options should be used to limit their impact.

The debate around the role of dispersal in outbreak dynamics is particularly pertinent to the management of the spruce budworm (hereafter SBW; *Choristoneura fumiferana* Clemens) (Lepidoptera: Tortricidae). The SBW is a univoltine native defoliator of balsam fir (*Abies balsamea* (L.) Mill) and spruce (*Picea spp.*) that periodically outbreaks (every ~35 years) and impacts large areas (> 10⁶ ha) of the boreal forest in North America (Royama, 1984). The severe economic consequences for forest industries and forestry-dependent communities

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(Chang et al., 2012) make it the most significant forest insect disturbance in North America's forests (Hardy et al., 1983). Dispersal by adult moths is common during outbreaks with moths exhibiting density-dependent dispersal such that high-density populations serve as source populations that export moths to low-density areas (Greenbank, 1963; Morris, 1963; Royama, 1984; Régnière and Nealis, 2007, 2019).

Historically, there are two interpretations of how SBW moth dispersal contributes to outbreaks. One interpretation favoured until recently is that moth dispersal functions mainly to synchronize outbreak cycles over large geographic areas (Régnière and Lysyk, 1995; Anderson and Sturtevant, 2011; Larroque et al., 2019; Régnière and Nealis, 2019). More recent evidence based in part on life table data suggests that dispersal may play a more complex role in outbreaks than previously thought, helping to synchronize outbreaks but also driving outbreak spread to outlying areas that might not otherwise reach outbreak densities (Régnière et al., 2019). This latter interpretation provides the basis for ongoing efforts to control outbreak spread in Atlantic Canada through detecting and controlling so-called 'hotspots' along the leading outbreak edge (Johns et al., 2019; MacLean et al., 2019).

A complementary approach to using demographic data to understand dispersal dynamics is to study spatial and temporal variations in large-scale spatial genetic structure (Lowe and Allendorf, 2010; Baguette et al., 2013). This approach is based on the premise that during the long endemic periods between outbreaks, areas with low population density experience limited genetic connectivity and become genetically differentiated (James et al., 2015). In this model, the endemic and isolated population is characterized by the genetic legacy of the previous outbreak collapse, whereas the epidemic and highly connected population is characterized by a spatial homogenization of multiple previous legacy populations at a spatial scale proportional to the outbreak stage and spatial extent. The consequent spatial and temporal genetic differentiation between sites from either successive outbreaks or in regions that are contemporaneously in different phases of the outbreak cycle can be thought of as a type of "isolation by outbreak cycle".

Fresh opportunities to test the impact of SBW dispersal on outbreak dynamics arose in 2006 with a new SBW outbreak on the north shore of the Saint Lawrence River in Quebec. Since then, the area affected has increased to > 9.6 million ha in Quebec (Ministère des Forêts, de la Faune et des Parcs [MFFP], 2019). Quebec could thus play the role of source population for adjacent jurisdictions in the current outbreak; dispersing moths from Quebec could trigger an outbreak in susceptible areas that have not yet been affected such as New Brunswick (Hennigar et al., 2013). Recently, Larroque et al. (2019) found a high level of genetic connectivity among Quebec SBW populations such that different outbreak patches in Quebec belong to the same genetic cluster, even though these sites are separated by distances up to one thousand kilometers. Despite these insights, there remain many questions around the degree of connectivity between this established, synchronous outbreak and the low-density endemic populations in New Brunswick and Maine. Part of the challenge in addressing these questions relates to the difficulty in obtaining SBW samples in low-density populations. Pheromone traps have been used to overcome this issue, but this approach raises a second problem, which is that captured moths are not necessarily local in provenance. More precise spatial genetic information would be gleaned if local populations were sampled prior to flight.

Here, we sought to study spatial genetic structure using larval SBW collected from within and outside of the leading outbreak edge in eastern Canada (Fig. 1). We specifically examined population genetic connectivity among the Canadian provinces of Quebec, New Brunswick, as well as adjacent Maine using spatial analysis of individual SBW larvae characterized using thousands of genome-wide SNPs (Larroque et al., 2019). A unique aspect of this study is that we were able to derive genetic information for difficult to sample low-density populations in New Brunswick and Maine through analysis of second instar larvae (L2s). Ongoing efforts at early intervention involve intensive

monitoring of SBW populations before defoliation occurs (MacLean et al., 2019). In this context, L2 larvae were sampled at ~1500 locations in New Brunswick and Maine in 2015 (MacLean et al., 2019). Our objectives were to: (1) develop a genetically informed approach to differentiate resident from migrant SBW, and (2) use this approach to determine the extent to which dispersing individuals from epidemic populations in Quebec are responsible for rising endemic populations.

We test the prediction that New Brunswick and Maine SBW populations are genetically distinct from the core of the outbreak in Quebec. Failure to confirm this prediction (*i.e.*, there is no differentiation between endemic and epidemic populations) would indicate that the spread of the ongoing outbreak depends on effective dispersal from epidemic to endemic regions. In contrast, if endemic and epidemic populations are found to be distinct, local population dynamics in the endemic region are more likely to be driving outbreak development and would suggest that dispersal plays a reduced role. We also consider a third, intermediate, possibility whereby genetic differentiation follows a continuous spatial gradient as a result of a travelling wave dynamic (Bjornstad et al., 1999). In the case of a travelling wave, one would expect genetic differentiation between epidemic and endemic SBW populations to be a function of temporally-lagged geographic distances between the two population types.

2. Materials and methods

2.1. Study area and sampling

The study uses samples collected in 2015 over a large area (237,154 km², Fig. 1) including Quebec, New Brunswick, and a part of Maine. In Quebec in the core of the current outbreak, late-instar (L6) spruce budworm larvae and pupae were collected over a 2-week period in June from nine locations ($n_{QC} = 148$; Fig. 1). Larvae and pupae were collected on live foliage, transported on branches in paper bags and were reared in the lab on synthetic diet until moth eclosion. Emerging moths were placed into 1.5 mL tubes and stored at -80°C until DNA extraction.

For New Brunswick and Maine, in the fall and early winter of 2015, 75 cm branches were collected from the mid-crown of spruce and fir trees at locations in New Brunswick and Maine and transported to a processing laboratory in Fredericton, New Brunswick. Overwintering L2 were extracted from branches using chemical extraction methods described in Miller et al. (1971). In total, 288 L2s ($n_{NewBrunswick} = 277$, $n_{Maine} = 11$) were collected and frozen at -80°C until DNA extraction.

2.2. DNA extraction

For the QC population, moths DNA was extracted from 12 to 20 individuals per site using the Qiagen DNA Blood and Tissue kit (Venlo, Limburg, NL) for a total of 148 individuals. DNA was extracted from 277 L2s in New Brunswick and 11 in Maine using modified NucleoSpin tissue XS kit (Macherey-Nagel) protocol. Samples were pre-lysed with 80 μL buffer B3 and 8 μL proteinase K solution, vortexed and spun in a mini centrifuge. Incubation was done at 56°C overnight. Lyse was completed by adding 80 μL buffer B3, vortexing briefly and incubating 5 min RT before adding 80 μL 96–100% EtOH, vortexing and spinning down. DNA binding was done by pipetting the supernatant into a column (2 mL collection tube + NucleoSpin tissue XS column). Samples were then centrifuged 1 min and washed twice by adding 50 μL Buffer B5 to the column and centrifuging 1 min the first time, 2 min the second time. DNA was then eluted with 15–20 μL buffer BE (incubation RT 5 min before centrifugation 1 min to elute). All centrifugations were done at 11,000g. Given the low sample size of the Maine population, we pooled both New Brunswick and Maine populations (hereafter called NB population, $n_{NB} = 288$).

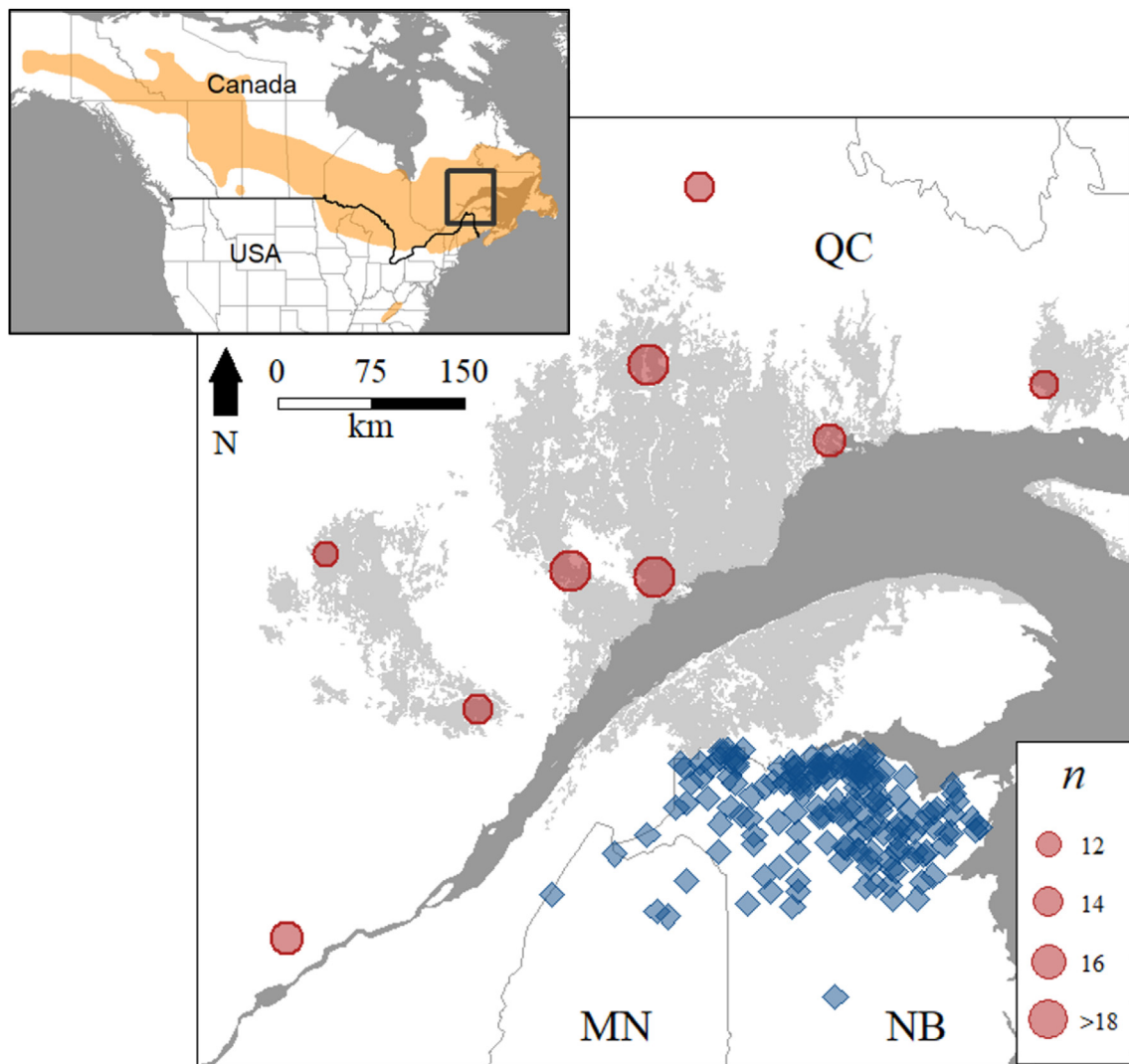


Fig. 1. Spruce budworm distribution range, study area, sampling locations and spruce budworm defoliation areas in 2015. Sites sampled in Quebec are represented by points whose size is proportional to the site's sample size (n). Individuals sampled in New Brunswick and Maine are represented by blue diamonds. Spruce budworm distribution range (adapted from Picq et al., 2018) is displayed in orange, and defoliation areas observed the year of the sampling (2015) are represented in grey.

2.3. Sequencing

All DNA samples were prepared for Genotyping-By-Sequencing (GBS, Elshire et al., 2011) using the methods described in Brunet et al. (2017). *PstI-MspI* GBS libraries (96 plex) were prepared by the Institut de Biologie Intégrative et des Systèmes (IBIS) at Université Laval (Quebec City, QC) using the protocol of Poland et al. (2012). Following final amplification of the pooled, adapter-ligated restriction fragments, 600 ng of each amplified library was normalized to remove the repetitive fraction by treatment with duplex-specific nuclease (Zhulidov et al., 2004). Finally, an additional PCR step using a selective reverse primer extending a single base (C) into the insert past the 3' restriction site was used to selectively amplify one-quarter of the total number of fragments, thereby increasing the read depth of sequenced fragments (Sonah et al., 2013). Single-end sequencing (100 bp reads) of these libraries was then performed with an Illumina HiSeq2000 (McGill University-Génome Québec Innovation Centre, Montreal, QC).

To detect the genetic variability specific to each population, bioinformatic processing of reads was performed separately for the L6/pupae data and the L2 data using the Fast-GBS pipeline (Torkamaneh et al., 2017). The Fast-GBS pipeline includes demultiplexing, trimming, mapping, and variant calling steps to process Genotyping-By-

Sequencing samples and provide highly accurate genotyping. This pipeline has been shown to yield the highest accuracy compared to the other pipelines (Torkamaneh et al., 2016). After demultiplexing and adapter trimming, reads less than 50 bp were discarded and remaining reads were aligned on the spruce budworm reference genome (bw6 version, Larroque et al., 2020) using the Burrows-Wheeler Aligner (BWA, Li and Durbin, 2010) with the *mem* algorithm. SNP calls were made as part of the Fast-GBS pipeline using Platypus (v0.8.1, Rimmer et al., 2014). Minimum read depth was set to eight reads. Only bi-allelic SNPs, with a maximum of 50% missing genotypes throughout all samples were retained. Individuals with more than 50% missing genotypes were removed. Finally, as some analytical methods that we use cannot handle missing data, missing genotypes were imputed using the software BEAGLE (v3.3.2, Browning and Browning, 2016) which replaces missing genotypes with the most frequently observed genotype associated with proximal SNP loci. Variants with R^2 (i.e., imputation accuracy) < 0.4 were removed as advised by Browning and Browning (2016).

2.4. Filtering SNP loci

SNPs with a minor allele frequency (MAF) < 5% were removed to

exclude putative sequencing errors and keep only the most informative SNPs (Marees et al., 2018). Similarly, SNPs in high linkage disequilibrium (LD) at a threshold of $r^2 \geq 0.2$ were discarded to remove highly correlated variants that add minimal extra information and could overly influence some methods (Price et al., 2008; Zou et al., 2010), using the *snpgdsLDpruning* function of the *SNPRelate* package (Zheng et al., 2012) in R (R Core Team, 2017).

When comparing NB and QC populations, we merged both datasets, keeping only shared loci. In the merged dataset, we removed SNPs showing deviation from expected Hardy-Weinberg equilibrium (HWE) as deviation from HWE can be indicative of genotyping errors (Anderson et al., 2010), null alleles (Brookfield, 1996) or selection (Wittke-Thompson et al., 2005). HWE was calculated using the *hw.test* function of the *pegas* package in R (Paradis, 2010) and using a Bonferroni correction for multiple comparisons. Finally, as our aim was to quantify the neutral evolutionary process of gene flow, we removed all SNPs identified as potentially under selection (Beaumont and Nichols, 1996) using the *pcadapt* function of the *pcadapt* package in R (Luu et al., 2017).

When focussing only on the NB population, we removed all SNPs showing deviation from HWE and loci potentially under selection for the reasons explained above and using the same procedures.

2.5. Genetic variation

If the growth of the NB population is independent from the massive outbreak in QC, we expect individuals to be differentiated into genetic groups that correspond to geographic provenance, with one group for the QC population (Larroque et al., 2019), and at least, one group for the NB population. The distribution of genetic variation among individuals was thus summarized with Principal Component Analysis (PCA) using the function *glPca* of the *ade4* package (Jombart et al., 2008) R package. We projected grouping factors (i.e., populations) on the 2-dimensional factorial map to assess whether the genetic differentiation matched with the spatial proximity of individuals.

2.6. Clustering

With the aim to explore the genetic structure across populations and to assess the source of putative immigrants, we searched for genetic groups using three approaches: (i) multivariate analysis through discriminant analysis of principal components (DAPC) (Jombart et al., 2010), (ii) population genetic model-based Bayesian clustering with the program STRUCTURE (Pritchard et al., 2000; Hubisz et al., 2009), and (iii) a least-squares optimisation approach with the R package TESS3 (Caye et al., 2016). All three methods endeavour to detect groups of genetically related individuals but differ in terms of their assumptions regarding population structure and the underlying evolutionary model.

We first used DAPC as implemented in the *ade4* package (Jombart, 2008) in R. DAPC maximizes differences among clusters while minimizing variation within and does not rely on a particular population genetics model, such as Hardy-Weinberg equilibrium. We applied the function *find.clusters* to determine the number of potential clusters K with $1 < K < 6$. Minimization of a Bayesian Information Criterion (BIC) was used to identify the most probable number of clusters present in the data.

STRUCTURE v.2.3.4 (Pritchard et al., 2000; Hubisz et al., 2009) was used assuming population admixture and correlated allele frequencies within populations. The Dirichlet parameter (λ), which describes how much the allele frequencies are correlated between populations, was inferred using the model of uncorrelated allele frequencies with $K = 1$ and its value was then used in the model of correlated allele frequencies ($\lambda = 0.744$). Simulations were run with 5×10^5 Markov Chain Monte Carlo (MCMC) iterations after a burn-in of 10^5 iterations. K was varied from 1 to 6 and 10 independent simulations were run for each K value to check for consistency in the results. Posterior probability ($\ln P(D)$),

was used to identify the most probable number of clusters (K) present in the data.

TESS3 is a spatially explicit ancestry estimation method implemented in the *tess3r* (Caye et al., 2016) package in R, that does not make assumptions concerning HWE or linkage disequilibrium. The number of ancestral populations (K) was evaluated using 20 independent runs of each K with $1 < K < 6$. A cross-validation procedure was used to select the best value of K according to the asymptote in the plot of cross-validation scores (Caye et al., 2016).

2.7. Isolation by distance

If the growth of the NB population is the result of dispersal of individuals from QC, then the genetic similarity between individuals in the two populations should be a function of the geographical distance between them. In this case, genetic and geographic distances are expected to be correlated (i.e., Isolation By Distance, IBD; Wright, 1943). We tested for IBD by testing the significance of the correlation between genetic distance and the geographic Euclidean distance between all pairs of individuals. We calculated inter-individual genetic distances using Principal Components Analysis (PCA). Significance of the correlation between the two distance matrices was assessed by carrying out a Mantel test using the *mantel.randtest* function of the *ade4* (Dray and Dufour, 2007) R package with 9999 permutations. We also assessed IBD within NB using the NB individuals' dataset containing all the loci selected in NB (4977 vs. 1000 shared loci between NB and QC). IBD within the QC population has been previously described (Larroque et al., 2019).

2.8. Spatial genetic structure

When populations exchange individuals they tend to become genetically similar. The integration of geographic and genetic information can improve our ability to identify weakly differentiated populations and can provide accurate spatial locations of cryptic clusters or genetic barriers (Storfer et al., 2007). Given the weak overall structure (i.e., clusters and IBD; see below), we tested for spatial genetic structure among QC and NB individuals, using spatial principal component analysis (sPCA; Jombart et al., 2008). This spatially-explicit multivariate method employs Moran's index (I) of spatial autocorrelation (Moran, 1948) to detect "global" structures, which correspond to positive spatial autocorrelation that may result from structuring of organisms in patches or clines (Jombart et al., 2008). We used the *sPCA* function implemented in the *ade4* package (Jombart et al., 2008) R package. We used the inverse distance method for weighting connectivity in the network. Significance was assessed using the permutation test ($n = 9999$) (Jombart et al., 2008). The individuals' scores of the first principal component were geographically mapped and interpolated to reveal spatial patterns of interest. Again, we also assessed "global structure" within NB using the NB individuals' dataset containing all the loci selected in NB (4977 vs. 1000 shared loci between NB and QC). Global structure within the QC population has been previously described (Larroque et al., 2019).

3. Results

3.1. Genotyping

Sequence processing through the fast-GBS pipeline (Torkamaneh et al., 2017) resulted in 73,960 and 93,215 high-quality SNPs for the QC population and the NB population respectively. These SNPs were identified using 140 L6/pupae and 160 L2s ($n_{\text{New Brunswick}} = 155$, $n_{\text{Maine}} = 5$) that satisfied the missing genotypes criterion ($< 50\%$). Most SNPs (73,489 for L6/pupae and 86,319 for L2) had at least one missing genotype which was then imputed, and 480 and one of the imputed SNPs were removed when filtering for R^2 for the L6/pupae and L2

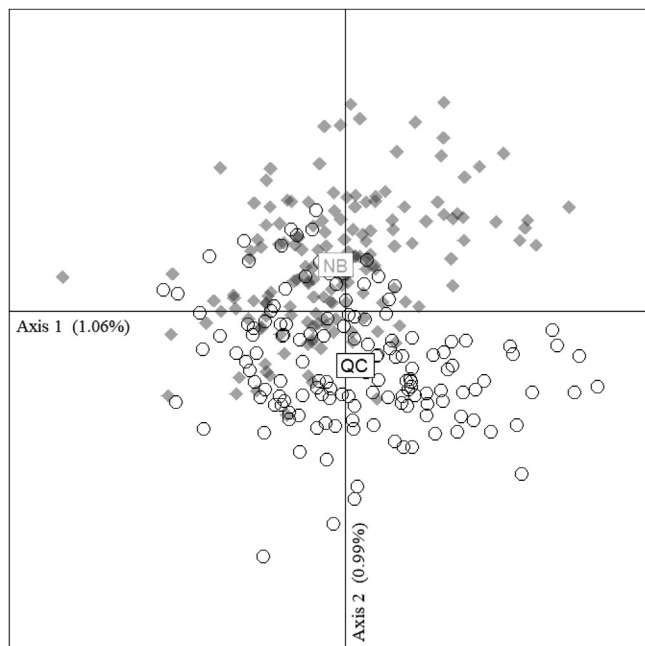


Fig. 2. Biplot of the principal components analysis of all individuals examined in QC and NB populations using 1000 filtered SNPs. QC and NB populations are displayed by circles and grey diamonds respectively. Percentage values associated with two first components (out of 300 components) represent the respective proportion of overall variance in allele frequencies captured.

respectively. Many SNPs (69,045 for L6/pupae and 85,475 for L2) were discarded when filtering for MAF and LD.

Of the remaining SNPs, 2708 deviated from HWE in NB population, and 54 of the remaining SNPs were detected as potentially under selection and removed. In total 4977 SNPs met our strict selection criteria and were retained for final analysis in NB population.

Of the 1178 shared loci between the L6/pupae and L2 data, 167 SNPs deviated from HWE, and 11 of the remaining SNPs were detected as potentially under selection and removed. In total 1000 SNPs met our strict selection criteria and were shared between the QC population and the NB population.

3.2. Genetic variation

The first two principal components (out of 300 components) accounted for 2.05% of the genotypic variation. The projection of populations onto PCA space revealed that QC and NB populations are not entirely genetically distinct (Fig. 2). While most of the QC and NB individuals are separated on the second principal component axis (Fig. 2; vertical axis), several individuals overlap, showing genetic similarity. Axis 2 is also associated with a North-South gradient in genetic variation ($F_{1,298} = 71.22$, $R^2 = 0.19$, $p < 10^{-3}$; Supporting information, Fig. S1).

3.3. Clustering

Neither the DAPC nor the STRUCTURE analysis was able to detect a significant genetic differentiation between the QC and NB populations. Concerning the DAPC analysis, a BIC minimum was identified at $K = 1$ suggesting the presence of a unique genetic cluster (Supporting information, Fig. S2A). No conclusive result was found with STRUCTURE, while an “atypical profile” of the posterior probability with the inferred number of K was obtained (Supporting information, Fig. S2B).

The algorithm for cross-validation curve of TESS3 revealed a likely number of $K = 2$ genetic clusters (Supporting information, Fig. S2C). Genetic difference between these two clusters was weak ($F_{st} = 0.004$)

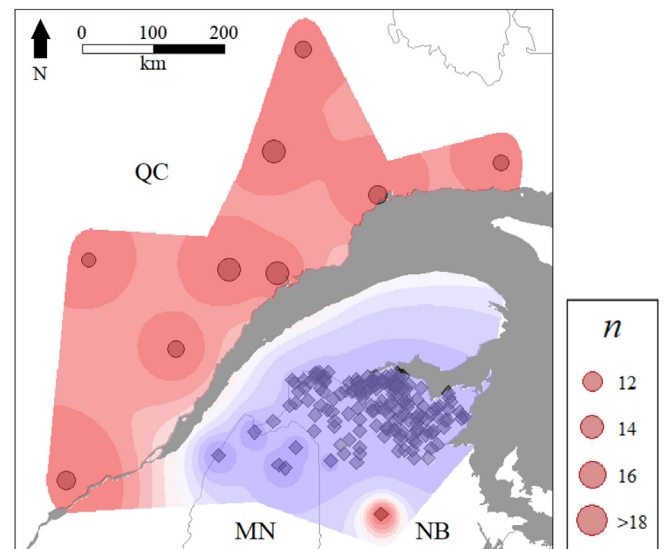


Fig. 3. Interpolated spatial genetic structure based on the individual ancestry coefficients of TESS3. The colour gradient indicates the degree of difference between individuals. Maximum differentiation is between dark blue and red. The extent of the interpolated region was defined as the concave hull polygon encompassing all the individuals sampling locations with an external buffer of 5 km. NB individuals sampling locations are represented by black diamonds and QC sampling sites are represented by points whose size is proportional to the site's sample size.

but significant ($p < 10^{-3}$). We spatially interpolated the ancestry coefficients projected them onto a map (Fig. 3). The resulting spatial pattern revealed that QC and NB populations are genetically distinct, with the notable exception of a single individual in far southern NB which was assigned to the QC cluster.

3.4. Isolation by distance

We found no evidence for significant individual-level IBD across populations ($p = 1$, Fig. 4A). Similarly, no IBD was detected when focussing on NB population only (i.e., using more loci; $p = 1$, Fig. 4B). The point density of pairwise distances for the NB population indicated a single large cluster of individuals in combination with a small group of individuals below the main cluster with the same level of genetic differentiation (i.e., genetic distance < 13 , Fig. 4B). We interpret this pattern to be indicative of a genetically distinct subset of individuals within NB.

3.5. Spatial genetic structure

We identified significant “global” spatial structure among QC and NB individuals ($p \leq 0.01$). The scores of the first positive eigenvalue have been interpolated and displayed on the area map with the maximal genetic differentiation between the blue and the red parts (Fig. 5A) and revealed the same pattern as that of TESS3, with QC and NB populations genetically distinct. A global spatial structure was also identified within the NB population only ($p = 0.03$) and scores of the first positive eigenvalue have been interpolated and displayed on the area map (Fig. 5B). Individuals exhibited a north-south gradient such that budworm closer to the Quebec border were more similar to the Quebec population and hence differentiated from individuals in the south of NB.

4. Discussion

Adult moth dispersal plays a major role in SBW outbreaks, synchronizing the dynamics of SBW populations over large geographic

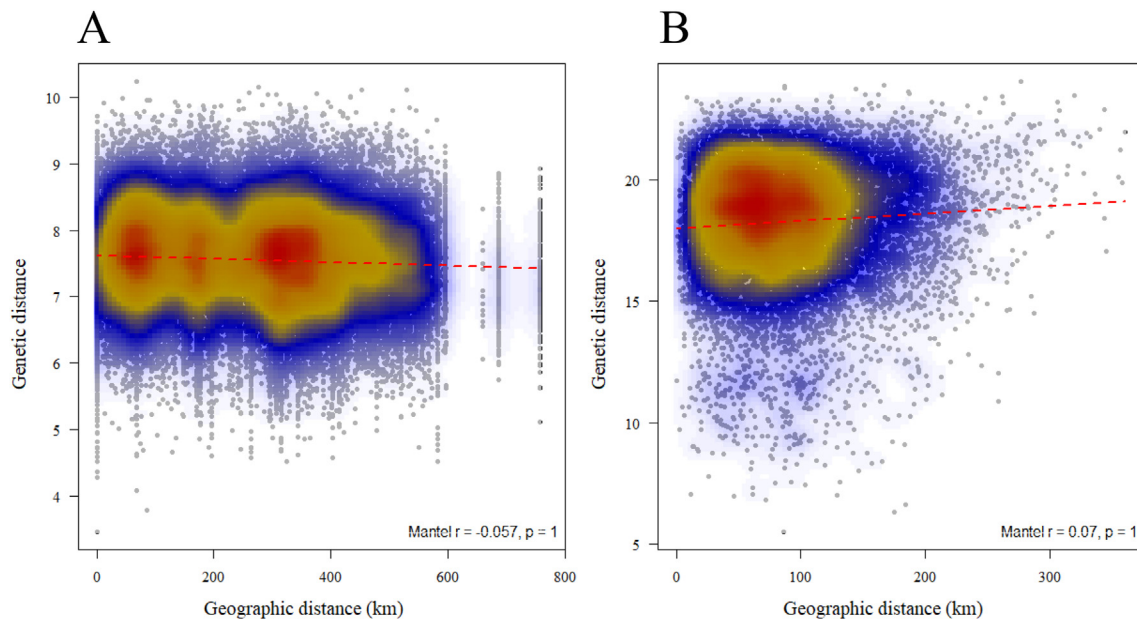


Fig. 4. Isolation by distance (IBD) plots. Scatterplots show the relationship between inter-individual genetic distances and geographic distances to test for the presence of IBD in (A) NB and QC, and (B) in NB only. Colours represent the relative density of points, with warmer colours indicating higher densities, while the dashed line shows the linear regression between the two distance matrices. The Mantel coefficient of correlation between geographic and genetic distances, as well as the associated p -value, are shown.

areas (Anderson and Sturtevant, 2011; Larroque et al., 2019; Régnière and Nealis, 2019). However, its contribution to the transition from the endemic to the epidemic level, *i.e.*, the respective contributions of local moths and migrants to local dynamics, is less clear. SBW dispersal is positively density-dependent, with source populations often exporting moths to low-density areas (Greenbank, 1963; Morris, 1963; Régnière and Nealis, 2007, 2019). Recent evidence based on recruitment data suggests that these dispersing moths may help to drive outbreak spread to outlying areas that might not otherwise reach outbreak densities (Régnière and Nealis, 2019). Yet, a better understanding of the role of dispersing moths in the initiation and maintenance of outbreaks is crucial for the development of management strategies that aim to mitigate consequences of outbreaks on both ecosystems and economies.

Using genetic data covering the core of the outbreak area in Quebec as well as outlying susceptible areas, we aimed to differentiate resident

from migrant SBW and identify the relative contribution of each to the rising endemic populations in New Brunswick and Maine. We found evidence of some genetic differentiation between Quebec and rising New Brunswick populations, as well as evidence of dispersal between them, particularly in the northern part of New Brunswick. These findings suggest that dispersing SBW moths play a significant role in the transition from the endemic to the epidemic state, with moths from Quebec potentially triggering the outbreak in adjacent jurisdictions. The existence of persisting spatial genetic structure within New Brunswick likely represents the spatial genetic legacy of the previous outbreak collapse (James et al., 2015) and suggests that local populations might not go extinct during endemic phases (Régnière and Nealis, 2019).

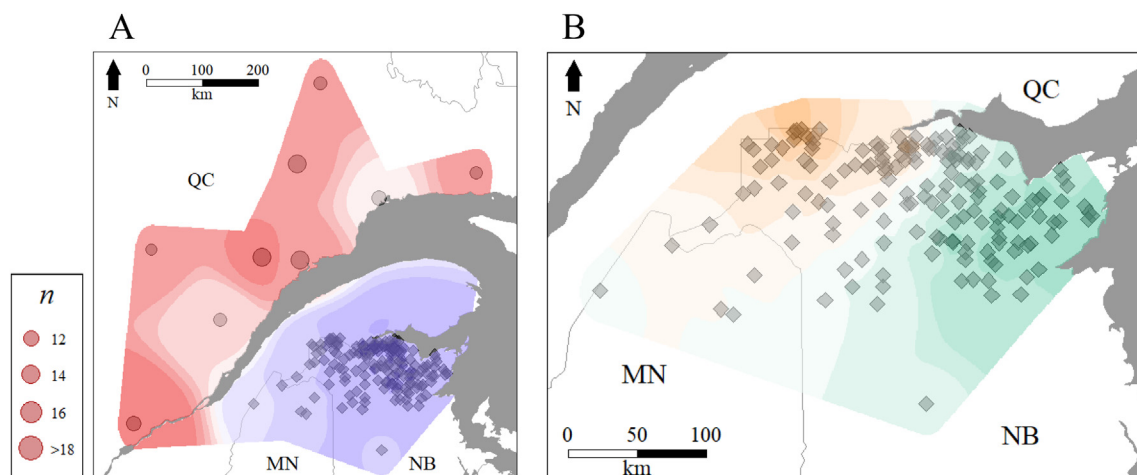


Fig. 5. Interpolated spatial genetic structure based on the first positive eigenvalue of the sPCA in (A) NB and QC, and (B) in NB only. The colour gradient indicates the degree of difference between individuals. Maximum differentiation is between (A) dark blue and red, and (B) orange and green. The extent of the interpolated region was defined as the concave hull polygon encompassing all the individuals sampling locations with an external buffer of 5 km. NB individuals sampling locations are represented by black diamonds and QC sampling sites are represented by points whose size is proportional to the site's sample size. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.1. Population genetic structure

SBW outbreaks occur every ~35 years. The long period of low population density between outbreaks, during which effective dispersal is reduced (Régnière and Nealis, 2007), is expected to produce detectable spatial genetic structure (James et al., 2015). Although the different clustering methods used in this study did not converge to the same solution, we found some evidence that suggests that Quebec and New Brunswick populations belong to two different genetic clusters (Fig. 3). Indeed, the DAPC and STRUCTURE did not find any clustering solution, but the TESS3 algorithm found two different genetic clusters that match the geographic origin of individuals, i.e., one cluster for the Quebec population and one cluster for the New Brunswick population.

The discrepancy among the three clustering methods used could be the result of the population genetics assumptions of STRUCTURE, such as Hardy-Weinberg equilibrium, which is unrealistic for outbreeding populations (Whitlock, 1992), and the inclusion of spatial information in TESS3 that yields increased power to detect low-level genetic structuring compared to the DAPC, as both methods do not rely on a particular population genetics model. The structure identified by TESS3 is also supported by the “global” structure detected on the plot of the sPCA, and the genetic differentiation between the two populations is low ($F_{st} = 0.004$) but significant ($p < 10^{-3}$). We were unfortunately unable to collect budworm samples in the lower St. Lawrence region of Quebec in 2015. However, a recent study demonstrated that in 2013, 2014 and 2015, all outbreeding populations sampled in Quebec belong to the same genetic cluster, including individuals from the Lower St. Lawrence region in 2014 (Larroque et al., 2019). That is, we found no evidence for genetic differentiation either spatially (at a scale of > 200 kms) or temporally (over the three years examined). As a result, we would not expect these sites to have changed from 2014 to 2015. Along with the lack of isolation by distance, these results suggest that both populations have been isolated from each other long enough during the endemic phase to have developed a weak genetic differentiation.

4.2. Traveling waves

In analysing New Brunswick separately from the epidemic population in Quebec, we were able to include more loci in our analysis and increase our statistical power to assess population genetic structure in this region. Despite this increased power, we did not find any indication of IBD, which suggests a high degree of population connectivity within New Brunswick. However, point densities in our IBD plot (Fig. 4B) revealed some individuals are genetically different from the main New Brunswick cluster. We also identified individuals within New Brunswick near the Quebec border that were distinct from the rest New Brunswick through our sPCA analysis (Fig. 5B). Finally, demographic survey data suggest increasing populations in areas near the Quebec border. In 2015, early instar larvae (L2) have been sampled each year at ~1500 locations in New Brunswick and Maine (MacLean et al., 2019). This survey identified hot spots of low, but rising populations close to the Quebec border. Whether these individuals are resident, migrants, or a combination of the two remains uncertain. However, these regions of increasing population conform with the areas of genetic differentiation.

Given these results, genetic differentiation between the two clusters within New Brunswick (Fig. 5B) appears to be the result of moth immigration from the Quebec outbreak cluster. The impact of dispersing moths from Quebec seem to be constrained to northern New Brunswick. Southern individuals in New Brunswick do not seem to be functionally connected to the outbreak in Quebec (Fig. 5). While being globally genetically distinct, a few individuals of both Quebec and New Brunswick showed genetic similarity (Fig. 2) and overall genetic variation exhibited a North-South structure (Supporting information, Fig. S1). Together, these results suggest that the southward spread of the current SBW outbreak is behaving as a travelling wave, that is, the process by which a population's peak density moves in one direction across a

spatial gradient over time (Bjornstad et al., 1999).

Traveling waves have been identified in several taxa (Liebhold et al., 2004) including other forest insect pests such as the larch budmoth (*Zeiraphera diniana*) in the European Alps (Johnson et al., 2004) or the autumnal moth (*Epirrita autumnata*) in northern Fennoscandia (Vindstad et al., 2019). Directional dispersal has been suggested as a potential explanation for travelling wave, especially for flying insect with long distance mass dispersal capacities (Bjornstad et al., 2002), characteristic of the SBW. During the current outbreak, the wave appears to have first reached the Quebec's south shore of the St. Lawrence River in 2012 (Supporting information, Fig. S3). At the time of sampling (2015), the wave reached the north of New Brunswick and seems to have stopped there, perhaps as a result of aggressive population management in New Brunswick (MacLean et al., 2019).

Essential to our interpretation of outbreak spread being driven by a travelling wave is the presence of spatial genetic structure within New Brunswick as a result of moth immigration (Fig. 5B). Although it is possible that this structure could have been present before the beginning of the outbreak, we think it is highly unlikely for the following three reasons: (1) we did not find any evidence of any barriers to dispersal within New Brunswick (i.e., no IBD; Fig. 4B); (2) mass dispersal events from Quebec in the Gaspésie and northern New Brunswick have been clearly demonstrated (Boulanger et al., 2017). The genetic structure we observe in New Brunswick corresponds strongly with the rising population density in New Brunswick in these areas that received high numbers of migrating individuals (Johns et al., 2019); And (3) we identified a significant North-South gradient associated with axis 2 of the PCA (Supporting information, Fig. S1) showing that New Brunswick individuals geographically close to Quebec are also genetically similar.

4.3. Conclusion

We found support for a genetic travelling wave pattern, matching the SBW density pattern. It thus seems that dispersal can synchronise the demography of distant populations during outbreaks (e.g., Larroque et al., 2019), but can also trigger the transition of populations from the endemic to the epidemic state by supplementing existing resident population (Régnière and Nealis, 2019). By supplementing endemic populations, migrants and particularly gravid female moths (Régnière and Nealis, 2019) may drive populations to beyond levels that can be controlled by local biotic constraints, thereby allowing populations to transition from an endemic to an epidemic state. This is in contrast to the hypothesis that compensatory dynamics (e.g., Allee effects) can result in effective local population extinction during endemic phases. If this were the case, one would expect to find a completely panmictic population with no signal from the New Brunswick population. These dynamics further highlight the complex patterns of spatial and temporal genetic diversity that can be expected as a result of repeated spatially synchronous population interruptions followed by asynchronous population collapse and genetic bottlenecks (James et al., 2015).

A key implication of this ‘spreading’ outbreak dynamic is that there may be an opportunity to contain the damage, providing that areas of spread can be both detected and controlled. This is the basis of ongoing management efforts in Atlantic Canada through the Early Intervention Strategy (Johns et al., 2019; MacLean et al., 2019). Further analyses of samples collected over large areas and as the outbreak develops could help to confirm the temporal stability of the observed genetic patterns and the extent to which management efforts are able to prevent further outbreak spread.

CRediT authorship contribution statement

Jeremy Larroque: Conceptualization, Methodology, Formal analysis, Writing - original draft. **Rob Johns:** Conceptualization, Writing - review & editing, Funding acquisition, Project administration. **Jade Canape:** Investigation, Formal analysis. **Benoit Morin:** Investigation,

Formal analysis. Patrick M.A. James: Conceptualization, Writing - review & editing, Funding acquisition, Supervision, Project administration.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2020.117965>.

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